



A globally convergent algorithm for lasso-penalized mixture of linear regression models

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ABSTRACT

Variable selection is an old and pervasive problem in regression analysis. One solution is to impose a lasso penalty to shrink parameter estimates toward zero and perform continuous model selection. The lasso-penalized mixture of linear regressions model (L-MLR) is a class of regularization methods for the model selection problem in the fixed number of variables setting. A new algorithm is proposed for the maximum penalized-likelihood estimation of the L-MLR model. This algorithm is constructed via the minorization–maximization algorithm paradigm. Such a construction allows for coordinate-wise updates of the parameter components, and produces globally convergent sequences of estimates that generate monotonic sequences of penalized log-likelihood values. These three features are missing in the previously presented approximate expectation–maximization algorithms. The previous difficulty in producing a globally convergent algorithm for the maximum penalized-likelihood estimation of the L-MLR model is due to the intractability of finding exact updates for the mixture model mixing proportions in the maximization-step. This issue is resolved by showing that it can be converted into a simple numerical root finding problem that is proven to have a unique solution. The method is tested in simulation and with an application to Major League Baseball salary data from the 1990s and the present day, where the concept of whether player salaries are associated with batting performance is investigated.

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1. Introduction

Variable selection is an old and pervasive problem in regression analysis and has been widely discussed because of this; see [George \(2000\)](#) and [Greene \(2003, Ch. 8\)](#) for classical introductions to the topic, and see [Hastie et al. \(2009, Ch. 3\)](#) and [Izenman \(2008, Ch. 5\)](#) for some modern perspectives. In recent years, regularization has become popular in the statistics and machine learning literature, stemming from the seminal paper of [Tibshirani \(1996\)](#) on the least absolute shrinkage and selection operator (lasso). A recent account of the literature regarding the lasso and related regularization methods can be found in [Bühlmann and van de Geer \(2011\)](#). The mixture of linear regressions (MLR) for modeling heterogeneous data was first considered in [Quandt \(1972\)](#). The introduction of the expectation–maximization (EM) algorithm by [Dempster et](#)

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al. (1977) made such models simpler to estimate in a practical setting. Subsequently, MLR models became more popular; see DeSarbo and Cron (1988), De Veaux (1989), and Jones and McLachlan (1992) for example.

The lasso-penalized MLR model (L-MLR) was considered in Khalili and Chen (2007) among a class of other regularization methods for the model selection problem in the fixed number of variables setting. The L-MLR was then generalized to the divergent number of variables setting in Khalili and Lin (2013), and to the mixture of experts setting in Khalili (2010). Furthermore, Stadler et al. (2010) (see also Bühlmann and van de Geer (2011, Sec. 9.2)) considered an alternative parameterization of the L-MLR to Khalili and Chen (2007), and suggested a modified regularization expression. An alternative modified grouped lasso criterion (Yuan and Lin, 2006) was suggested for regularization of the MLR model in Hui et al. (2015). A recent review of the literature regarding the variable selection problem in MLR models can be found in Khalili (2011).

In this article, we propose a new algorithm for the maximum penalized-likelihood (MPL) estimation of L-MLR models. This algorithm is constructed via the MM (minorization–maximization) algorithm paradigm of Lange (2013, Ch. 8). Such a construction allows for some desirable features such as coordinate-wise updates of parameters, monotonicity of the penalized likelihood sequence, and global convergence of the estimates to a stationary point of the penalized log-likelihood function. These three features are missing in the approximate-EM algorithm presented in Khalili and Chen (2007). Previously, MM algorithms have been suggested for the regularization of regression models in Hunter and Li (2005), where they are noted to be numerically stable. Coordinate-wise updates of parameters in lasso-type problems were considered in Wu and Lange (2008), who also noted such updates to be fast and stable when compared to alternative algorithms. Furthermore, Stadler et al. (2010) also consider a coordinate-wise update scheme in their generalized EM algorithm, although the global convergence properties of the algorithm could only be established for the MPL estimation of a modified case of the L-MLR model with a simplified penalization function. Note that we use the phrase “global convergence” to mean that the algorithm generates a sequence of updates that converge to some limiting value, when initialized from any point in the parameter space. This is in opposition to the alternative interpretation that the algorithm generates a sequence of updates that converge to a globally optimal solution of the problem being solved. Our use of the phrase “global convergence” is in compliance with the terminology used in the optimization literature; see Sriperumbudur and Lanckriet (2012) for example.

The difficulty in producing a globally convergent algorithm for the MPL estimation of the L-MLR model, which led both Khalili and Chen (2007) and Stadler et al. (2010) to utilize approximation schemes, is due to the intractability of the problem of updating the mixture model mixing proportions in the maximization-step of their respective algorithms. In our algorithm, we solve this issue by showing that it can be converted into a simple numerical root finding problem and prove that a unique solution exists. Aside from the new algorithm, we also consider the use of the L-MLR as a screening mechanism in a two-step procedure, as suggested in Bühlmann and van de Geer (2011, Sec. 2.5). Here, the L-MLR model is used to select the variable subset (step one) to include in a subsequent estimation of a MLR model (step two). This procedure allows for the use of generic asymptotic results for quasi-maximum likelihood estimation, such as those of Amemiya (1985); see also White (1982). Optimization of the lasso tuning parameter vector λ via derivative free numerical methods is also explored as an alternative to exhaustive grid search.

To supplement the presented algorithm (see Appendix A) and procedures, we perform a set of simulation studies to demonstrate the capacity of our methodology. A user-friendly program that implements the proposed algorithm in C++ is also available at <https://github.com/lukelloydjones/LMLR> and is shown to be capable of handling reasonably large estimation problems. To compare the performance of the method on real data, we analyze the same data set on Major League Baseball (MLB) salaries presented in Khalili and Chen (2007). This allows for an initial comparison with the foundational work and an exploration of whether common measures of batting performance are good predictors of how much a batter is paid. This analysis is supplemented with a current data set from MLB seasons 2011–15, which allows for an investigation into how the distribution of salaries has changed and whether the same or new predictors are relevant. Baseball has always had a fascination with statistics, with baseball's link with statistics going back to the origins of the sport (Marchi and Albert, 2013). In particular, economists have long had great interest in the labor market and finances associated with MLB (Brown et al., 2015). Of notable fame is the Moneyball hypothesis (Lewis, 2004), which states that the ability of a player to get ‘on base’ was undervalued in the baseball labor market (before 2003) (Hakes and Sauer, 2006). Baseball statistics on player and team performance are some of the best kept of any sport especially in the modern era. Furthermore, baseball owners and players agree that playing performance is measurable and is associated with salary (Scully, 1974; Fullerton Jr. and Peach, 2016). In this article we emphasize the application of our new methodology to these data in so far as there exists a statistical association, and believe that the implications of the performance salary association, with respect to MLB as a whole, have been explored in more depth elsewhere.

The article proceeds as follows. In Section 2, we introduce the L-MLR model and present the MM algorithm for its MPL estimation. In Section 3, we discuss the use of L-MLR models for statistical inference, and present the two-stage screening and estimation procedure. Section 4 outlines the algorithm's implementation. Simulation studies are then presented in Section 5, and we apply our method to data from salaries of batters from Major League Baseball (MLB) from the 1990s and present day in Section 6. Conclusions are then drawn in Section 7.

2. Mixture of linear regressions model

Let $Y_1, \dots, Y_n \in \mathbb{R}$ be an independent and identically distributed (IID) random sample that is dependent on corresponding covariate vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$, and let Z_t be a latent categorical random variable ($t = 1, \dots, n$) such that $z_t \in \{1, \dots, g\}$, where $\mathbb{P}(Z_t = i) = \pi_i > 0$ and $\sum_{i=1}^g \pi_i = 1$. The MLR model can be defined via the conditional probability density characterization

$$f(y_t | \mathbf{x}_t, Z_t = i; \boldsymbol{\theta}) = \phi(y_t; \alpha_i + \mathbf{x}_t^T \boldsymbol{\beta}_i, \sigma_i^2),$$

which implies the marginal probability density characterization

$$f(y_t | \mathbf{x}_t; \boldsymbol{\theta}) = \sum_{i=1}^g \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \boldsymbol{\beta}_i, \sigma_i^2). \quad (1)$$

Here $\phi(y; \mu, \sigma^2)$ is a normal density function with mean μ and variance σ^2 , and we say that $\phi(y_t; \alpha_i + \mathbf{x}_t^T \boldsymbol{\beta}_i, \sigma_i^2)$ is the i th mixture component density. The vectors $\boldsymbol{\beta}_i = (\beta_{i1}, \dots, \beta_{ip})^T \in \mathbb{R}^p$, and scalars $\alpha_i \in \mathbb{R}$ and $\sigma_i^2 > 0$ are the specific regression coefficients, intercepts, and variances of the i th component density, respectively. We put all of the parameter components into the parameter vector $\boldsymbol{\theta} = (\boldsymbol{\pi}^T, \boldsymbol{\alpha}^T, \boldsymbol{\beta}^T, \boldsymbol{\sigma}^T)^T$, where $\boldsymbol{\pi} = (\pi_1, \dots, \pi_g)^T$, $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_g)^T$, $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^T, \dots, \boldsymbol{\beta}_g^T)^T$, and $\boldsymbol{\sigma} = (\sigma_1^2, \dots, \sigma_g^2)^T$.

We let $y_1, \dots, y_n$ with covariates $\mathbf{x}_1, \dots, \mathbf{x}_n$ be an observed random sample arising from an MLR with unknown parameter $\boldsymbol{\theta}_0 = (\boldsymbol{\pi}_0^T, \boldsymbol{\alpha}_0^T, \boldsymbol{\beta}_0^T, \boldsymbol{\sigma}_0^T)^T$. If no additional assumptions are made regarding the nature of the regression coefficients $\boldsymbol{\beta}_0$ the parameter vector can be estimated by the ML estimator $\tilde{\boldsymbol{\theta}}_n$, where $\tilde{\boldsymbol{\theta}}_n$ is an appropriate local maximizer of the log-likelihood function for the MLR model

$$\mathcal{L}_n(\boldsymbol{\theta}) = \sum_{t=1}^n \log \sum_{i=1}^g \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \boldsymbol{\beta}_i, \sigma_i^2).$$

2.1. Lasso-penalized MLR

Suppose that it is known that $\boldsymbol{\beta}_0$ is sparse, in the sense that some or many elements of $\boldsymbol{\beta}_0$ are exactly equal to zero. The estimates for the zero elements of $\boldsymbol{\beta}_0$, obtained via $\tilde{\boldsymbol{\theta}}_n$, will tend to be close to zero but will not be shrunk exactly to zero, and thus cannot be completely excluded from the model without the use of some other elimination techniques, such as hypothesis testing. One method for simultaneously shrinking insignificant regression coefficients to zero and estimating the parameter vector $\boldsymbol{\theta}_0$, as suggested by [Khalili and Chen \(2007\)](#), is to estimate the L-MLR by computing the MPL estimator $\hat{\boldsymbol{\theta}}_n$, where $\hat{\boldsymbol{\theta}}_n$ is an appropriate local maximizer of the lasso-penalized log-likelihood function for the MLR model

$$\mathcal{F}_n(\boldsymbol{\theta}) = \mathcal{L}_n(\boldsymbol{\theta}) - \mathcal{P}_n(\boldsymbol{\theta}). \quad (2)$$

Here

$$\mathcal{P}_n(\boldsymbol{\theta}) = \sum_{i=1}^g \pi_i \sum_{j=1}^p \lambda_{in} |\beta_{ij}| \quad (3)$$

is the mixture lasso penalty function, where $\lambda_{in} = n^{1/2} \gamma_{in}$ and $\gamma_{in} \geq 0$ are sequences of penalizing constants that can be set to obtain a desired level of sparsity in the model. We note that $\tilde{\boldsymbol{\theta}}_n$ and $\hat{\boldsymbol{\theta}}_n$ are equivalent if $\lambda_{in} = 0$ for each i .

We now proceed to construct an MM algorithm for the MPL estimation of the L-MLR model. In order to produce an algorithm that is globally convergent, we follow the tactic of [Hunter and Li \(2005\)](#) and consider instead an appropriate local maximizer to the ε -approximate lasso-penalized log-likelihood function

$$\mathcal{F}_{n,\varepsilon}(\boldsymbol{\theta}) = \mathcal{L}_n(\boldsymbol{\theta}) - \mathcal{P}_{n,\varepsilon}(\boldsymbol{\theta}), \quad (4)$$

where

$$\mathcal{P}_{n,\varepsilon}(\boldsymbol{\theta}) = \sum_{i=1}^g \pi_i \sum_{j=1}^p \lambda_{in} \sqrt{\beta_{ij}^2 + \varepsilon^2} \quad (5)$$

for some small $\varepsilon > 0$. Similarly to [Hunter and Li \(2005, Prop. 3.2\)](#), we can show that $|\mathcal{F}_{n,\varepsilon}(\boldsymbol{\theta}) - \mathcal{F}_n(\boldsymbol{\theta})| \rightarrow 0$ uniformly as $\varepsilon \rightarrow 0$, over any compact subset of the parameter space. The analysis of (4) instead of (2) is advantageous since its differentiability allows for the simple application of a useful global convergence theorem.

Remark 1. In general when conducting parameter estimation of a parametric model, be it via ML estimator or MPL estimation, one must also consider whether or not the obtained parameters determine the underlying data generating process. This is the general problem of identifiability, in parametric statistics. In the case of MLR models of form (1), the problem of identifiability has been considered by Jiang and Tanner (1999) and Hennig (2000), and more recently in Compiani and Kitamura (2016). Since the usual definition of one-to-one mapping between parameter and model cannot be applied directly to mixture models (i.e., due to the equivalence of probabilities up to a permutation of the parameter vector components), other notions must be used instead. In Jiang and Tanner (1999), the notion of initialized-and-ordered identifiability is introduced, whereupon one identifies the parameters of mixtures by setting the g th component mixture to have probability $\pi_g = 1 - \sum_{i=1}^{g-1} \pi_i$ and ordering the permutable component elements of θ via the natural lexicographical ordering (cf. Khan et al., 2015, Example 2.1.10). Lemma 3 and Theorem 1 of Jiang and Tanner (1999) prove that model (1) is initialized-and-ordered identifiable. We note that the definition of Jiang and Tanner (1999) is appropriate for use in large data situations, such as the ones where regularization is needed, where there are many uniquely observed covariate vectors $\mathbf{X}_t = \mathbf{x}_t$. When the number of unique covariate vectors in the data is small such that multiplicity becomes an issue (e.g., in the sense of Breiman et al. (2001)), a more appropriate mode of identifiability may be that of Hennig (2000), which assumes that the covariate values $\mathbf{X}_t = \mathbf{x}_t$ are fixed, and studies the interaction between the covariates $\mathbf{x}_i$ and the parameter vector θ . In contrast, individual values of the covariates can be ignored under the identifiability framework of Jiang and Tanner (1999), as $\mathbf{X}_t$ is assumed to be a random distribution over a subset of the Euclidean space with nonnull interior, and identifiability is considered from the perspective of the joint density functions on $(\mathbf{X}_t^T, Y_t)$. Hennig (2000), however, considers identifiability from the perspective of the conditional density functions on $Y_t | \mathbf{X}_t = \mathbf{x}_t$. We further note that to the best of our knowledge, there has been no direct treatment of the difference in identifiability of model (1) when estimated with or without variable selection.

2.2. Minorization–maximization algorithm

We shall now proceed to describe the general framework of the block-wise MM algorithm. We note that these algorithms are also known as block successive lower-bound maximization (BSLM) algorithms in the language of Razaviyayn et al. (2013).

Suppose that we wish to maximize some objective function $\mathcal{F}(\theta)$, where $\theta = (\theta_1^T, \dots, \theta_m^T)^T \in \Theta \subset \mathbb{R}^q$ and $\theta_k \in \Theta_k \subset \mathbb{R}^{q_k}$, for $k = 1, \dots, m$ and $\sum_{k=1}^m q_k = q$. If $\mathcal{F}(\theta)$ is difficult to maximize (e.g., the first order conditions of $\mathcal{F}$ are intractable, or $\mathcal{F}$ is non-differentiable), then we seek to replace its direct maximization by a sequence of simpler problems that are manageable and solvable via available techniques.

We say that $\mathcal{U}_k(\theta_k; \psi)$ is a minorizer of $\mathcal{F}$ over coordinate block k if $\mathcal{F}(\psi) = \mathcal{U}_k(\psi_k; \psi)$ and $\mathcal{F}(\psi) \geq \mathcal{U}_k(\psi_1, \dots, \psi_{(k-1)}, \theta_k, \psi_{(k+1)}, \dots, \psi_m)$ whenever $\theta_k \neq \psi_k$. Here, $\psi \in \Theta$ and $\psi_k \in \Theta_k$. When $m = 1$, we say that $\mathcal{U}_1(\theta_1; \psi) = \mathcal{U}(\theta; \psi)$ minorizes $\mathcal{F}$. Upon finding an appropriate set of block-wise minorizers of $\mathcal{F}$, we can define a block-wise MM algorithm as follows.

Let $\theta^{(0)}$ be some initialization and let $\theta^{(r)}$ be the r th iterate of the algorithm. In the $(r + 1)$ th iteration, block-wise MM update scheme proceeds by computing

$$\theta_k^{(r+1)} = \arg \max_{\theta_k \in \Theta_k} \mathcal{U}_k(\theta_k; \theta^{(r)})$$

for $k = (r \bmod m) + 1$, and setting $\theta_l^{(r+1)} = \theta_l^{(r)}$ for all $l \neq k$, where $l = 1, \dots, m$. Given the definition of block-wise minorizers, it is not difficult to see that the iteration scheme that is described implies the monotonic ascent property

$$\mathcal{F}(\theta^{(r+1)}) \geq \mathcal{U}_k(\theta_k^{(r+1)}; \theta^{(r)}) \geq \mathcal{U}_k(\theta_k^{(r)}; \theta^{(r)}) = \mathcal{F}(\theta^{(r)})$$

for every coordinate block k . We now present a set of useful minorizers for the MPL estimation of the L-MLR model.

Lemma 1 (Minorizers). If $\theta, \varepsilon \in \mathbb{R}$, then the function $\mathcal{F}(\theta) = -\sqrt{\theta^2 + \varepsilon^2}$ can be minorized by

$$\mathcal{U}(\theta; \psi) = -\frac{\theta^2}{2w(\psi)} + C(\psi), \quad (6)$$

where $C(\psi) = -2^{-1}w(\psi) - \varepsilon^2 w(\psi)^{-1}$, and $w(\psi) = \sqrt{\psi^2 + \varepsilon^2}$.

If $\Theta = [0, \infty)^q$, then the function $\mathcal{F}(\theta) = \log(\sum_{i=1}^q \theta_i)$ can be minorized by the function

$$\mathcal{U}(\theta; \psi) = \sum_{i=1}^q \tau_i(\psi) \log \theta_i + C(\psi), \quad (7)$$

where $\tau_i(\psi) = \psi_i / \sum_{j=1}^q \psi_j$ and $C(\psi) = -\sum_{i=1}^q \tau_i(\psi) \log \tau_i(\psi)$.

If $\theta, \mathbf{x} \in \mathbb{R}^q$ and $\alpha, y \in \mathbb{R}$, then the function $\mathcal{F}(\theta) = -(y - \alpha - \mathbf{x}^T \theta)^2$ can be minorized by

$$\mathcal{U}(\theta; \psi) = \frac{1}{q} \sum_{j=1}^q -[y - \alpha - q x_j (\theta_j - \psi_j) - \mathbf{x}^T \psi]^2. \quad (8)$$

Minorizers (6) and (8) of Lemma 1 are adapted from the minorizers presented in Lange (2013, Sec. 8.5), and minorizer (7) can be found in Zhou and Lange (2010).

2.3. Derivation of minorizers for the maximization of $\mathcal{F}_{n,\varepsilon}(\theta)$

Conditioned on the r th iterate $\theta^{(r)}$, we can minorize $-\mathcal{P}_{n,\varepsilon}(\theta)$ by

$$\mathcal{U}_1(\theta; \theta^{(r)}) = -\frac{1}{2} \sum_{i=1}^g \pi_i \sum_{j=1}^p \lambda_{ij} \frac{\beta_{ij}^2}{w_{ij}^{(r)}} + C_1(\theta^{(r)}), \quad (9)$$

where $w_{ij}^{(r)} = \sqrt{\beta_{ij}^{(r)2} + \varepsilon^2}$ and

$$C_1(\theta^{(r)}) = -\frac{1}{2} \sum_{i=1}^g \pi_i \sum_{j=1}^p \lambda_{ij} w_{ij}^{(r)} - \frac{\varepsilon^2}{2} \sum_{i=1}^g \pi_i \sum_{j=1}^p \lambda_{ij} \frac{1}{w_{ij}^{(r)}},$$

by making the substitution $\theta = \beta_{ij}$ in (6) of Lemma 1, for each i and j . We can write $\mathcal{U}_1(\theta; \theta^{(r)}) = \mathcal{U}_1(\pi, \beta; \theta^{(r)})$ since it does not depend on α or σ . Next, we can minorize $\mathcal{L}_n(\theta)$ by

$$\begin{aligned} \mathcal{U}_2(\theta; \theta^{(r)}) &= \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} [\log \pi_i + \log \phi(y_t; \alpha_i + \mathbf{x}_t^T \beta_i, \sigma_i^2)] - \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \tau_{it}^{(r)} \\ &= \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \pi_i - \frac{1}{2} \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \sigma_i^2 \\ &\quad - \frac{1}{2} \sum_{i=1}^g \sum_{t=1}^n \frac{\tau_{it}^{(r)}}{\sigma_i^2} (y_t - \alpha_i - \mathbf{x}_t^T \beta_i)^2 + C_2(\theta^{(r)}) \end{aligned} \quad (10)$$

where $\tau_{it}^{(r)} = \pi_i^{(r)} \phi(y_t; \alpha_i^{(r)} + \mathbf{x}_t^T \beta_i^{(r)}, \sigma_i^{2(r)}) / f(y_i | \mathbf{x}_t; \theta^{(r)})$. Note that $\tau_{it}^{(r)}$ and $\mathcal{U}_2(\theta; \theta^{(r)})$ are analogous to the posterior probability and the expected complete-data log-likelihood function of the expectation-maximization algorithm respectively. In (10),

$$C_2(\theta^{(r)}) = -\frac{n}{2} \log(2\pi) - \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \tau_{it}^{(r)},$$

by making the substitution $\theta_i = \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \beta_i, \sigma_i^2)$ in (7) for each i . We can further minorize $\mathcal{U}_2(\theta; \theta^{(r)})$ by

$$\begin{aligned} \tilde{\mathcal{U}}_2(\theta; \theta^{(r)}) &= \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \pi_i - \frac{1}{2} \sum_{i=1}^g \sum_{t=1}^n \tau_{it}^{(r)} \log \sigma_i^2 \\ &\quad - \frac{1}{2p} \sum_{i=1}^g \sum_{j=1}^p \sum_{t=1}^n \frac{\tau_{it}^{(r)}}{\sigma_i^2} \left(y_t - \alpha_i - p x_{tj} (\beta_{ij} - \beta_{ij}^{(r)}) - \mathbf{x}_t^T \beta_i^{(r)} \right)^2 \\ &\quad + C_2(\theta^{(r)}), \end{aligned} \quad (11)$$

by making the substitutions $\alpha = \alpha_i$, $y = y_t$, $\mathbf{x} = \mathbf{x}_t$, and $\theta = \beta_i$ in (8), for each i, j , and t .

Using results (9)–(11), we can deduce the following block-wise minorizers for the ε -approximate lasso-penalized log-likelihood function.

Proposition 1. Conditioned on the r th iterate $\theta^{(r)}$, (4) can be block-wise minorized in the coordinates of the parameter components π , α and σ , and β , via the minorizers

$$\mathcal{U}_\pi(\pi; \theta^{(r)}) = \mathcal{U}_2(\pi, \alpha^{(r)}, \beta^{(r)}, \sigma^{(r)}; \theta^{(r)}) - \mathcal{P}_{n,\varepsilon}(\pi, \beta^{(r)}), \quad (12)$$

$$\mathcal{U}_{\alpha,\sigma}(\alpha, \sigma; \theta^{(r)}) = \mathcal{U}_2(\pi^{(r)}, \alpha, \beta^{(r)}, \sigma; \theta^{(r)}) - \mathcal{P}_{n,\varepsilon}(\theta^{(r)}), \quad (13)$$

and

$$\mathcal{U}_\beta(\boldsymbol{\beta}; \boldsymbol{\theta}^{(r)}) = \mathcal{U}_1(\boldsymbol{\pi}^{(r)}, \boldsymbol{\beta}; \boldsymbol{\theta}^{(r)}) + \tilde{\mathcal{U}}_2(\boldsymbol{\pi}^{(r)}, \boldsymbol{\alpha}^{(r)}, \boldsymbol{\beta}, \boldsymbol{\sigma}^{(r)}; \boldsymbol{\theta}^{(r)}), \quad (14)$$

respectively.

2.4. Maximization of minorizers

We now seek to maximize (12) in the constraint set

$$\tilde{\Theta}_\pi = \left\{ \boldsymbol{\pi} : \pi_i > 0, \sum_{i=1}^g \pi_i = 1, i = 1, \dots, g \right\}. \quad (15)$$

This can be achieved by solving for the appropriate root of the FOC (first-order condition) for the Lagrangian

$$\Lambda(\boldsymbol{\pi}, \zeta) = \mathcal{U}_\pi(\boldsymbol{\pi}, \boldsymbol{\theta}^{(r)}) - \zeta \left(\sum_{i=1}^g \pi_i - 1 \right), \quad (16)$$

which enforces the equality constraint from (15). The appropriate root in this case is the one that also satisfies the inequality constraints from (15).

Let ∇ denote the gradient operator. The FOC, in the case of (16), is $\nabla \Lambda = \mathbf{0}$, where $\mathbf{0}$ is the zero vector. The FOC reduces to the set of simultaneous equations

$$a_i/\pi_i - b_i - \zeta = 0, \quad (17)$$

for $i = 1, \dots, g$, and $\sum_{i=1}^g \pi_i - 1 = 0$. Here, $a_i = \sum_{t=1}^n \tau_{it}^{(r)}$ and $b_i = \lambda_{in} \sum_{j=1}^p \left| \beta_{ij}^{(r)} \right|$, for each i .

Note that we can write (17) as $\pi_i = a_i/(\zeta + b_i)$, for each i . Write $h(\boldsymbol{\pi}) = \sum_{i=1}^g \pi_i$ and substitute in (17) to obtain

$$h(\zeta) = \sum_{i=1}^g \frac{a_i}{\zeta + b_i}. \quad (18)$$

By taking the derivative of (18), we obtain $dh/d\zeta = -\sum_{i=1}^g a_i/(\zeta + b_i)^2 < 0$, for any valid input ζ . Thus (18) is strictly decreasing over any interval for which it is defined. Next, notice that (18) is continuous, and that $\lim_{\zeta \rightarrow \infty} h(\zeta) = 0$ and $\lim_{\zeta \rightarrow b^*} h(\zeta) = \infty$, where $b^* = -\min_{i=1, \dots, g} \{b_i\}$. Therefore, the equation

$$h(\zeta) - 1 = 0 \quad (19)$$

must have a unique solution in the open interval (b^*, ∞) , by the intermediate value theorem. Thus, an appropriate root to the FOC can be found by numerically searching for a solution within this interval and substituting the solution, say ζ^* , into (17) to obtain the estimates $\pi_i^* = a_i/(\zeta^* + b_i)$, for each i .

Therefore, to summarize, we can compute the $(r + 1)$ th iterate block-wise update

$$\boldsymbol{\pi}^{(r+1)} = \arg \max_{\boldsymbol{\pi} \in \Theta_\pi} \mathcal{U}_\pi(\boldsymbol{\pi}; \boldsymbol{\theta}^{(r)}) \quad (20)$$

by setting $\pi_i^{(r+1)} = a_i/(\zeta^* + b_i)$, for each i , where ζ^* is the unique root of (19) in the interval (b^*, ∞) .

Remark 2. Note that there are potentially other roots of equation $h(\zeta) - 1 = 0$ in each of the intervals $(b_{(i-1)}, b_{(i)})$, where $-\infty = b_{(0)} < b_{(1)} < \dots < b_{(g)}$, for $i = 1, \dots, g$. Here, $b_{(i)}$ is the i th order statistic of the set $\{b_1, \dots, b_g\}$. However, if ζ takes a value less than b^* , then at least one of the component proportions π_i will be less than 0, which violates the constraint set (15). This is because $a_i > 0$ and $b_i > 0$, for each i . Thus, the only appropriate root is in the interval (b^*, ∞) .

Fortunately, the block-wise updates for $\boldsymbol{\alpha}$ and $\boldsymbol{\sigma}$ can be obtained from (13) via the first-order condition $\nabla \mathcal{U}_{\boldsymbol{\alpha}, \boldsymbol{\sigma}}(\boldsymbol{\alpha}, \boldsymbol{\sigma}; \boldsymbol{\theta}^{(r)}) = \mathbf{0}$. By doing so, we obtain the updates

$$\alpha_i^{(r+1)} = \frac{\sum_{t=1}^n \tau_{it}^{(r)} \left(y_t - \sum_{j=1}^p \beta_{ij}^{(r)} x_{tj} \right)}{\sum_{t=1}^n \tau_{it}^{(r)}} \quad (21)$$

and

$$\sigma_i^{2(r+1)} = \frac{\sum_{t=1}^n \tau_{it}^{(r)} \left(y_t - \alpha_i^{(r+1)} - \sum_{j=1}^p \beta_{ij}^{(r)} x_{tj} \right)^2}{\sum_{t=1}^n \tau_{it}^{(r)}} \quad (22)$$

for each i . Similarly, solving $\nabla \mathcal{U}_\beta(\boldsymbol{\beta}; \boldsymbol{\theta}^{(r)}) = \mathbf{0}$ yields the coordinate-wise updates for the $\boldsymbol{\beta}$ block

$$\beta_{ij}^{(r+1)} = \frac{p \sum_{t=1}^n \tau_{it}^{(r)} x_{ij}^2 \beta_{ij}^{(r)} + \sum_{t=1}^n \tau_{it}^{(r)} x_{ij} \left(y_t - \alpha_i^{(r)} - \sum_{j=1}^p \beta_{ij}^{(r)} x_{ij} \right)}{\sigma_i^{2(r)} \pi_i^{(r)} \lambda_{in}/w_{ij}^{(r)} + p \sum_{t=1}^n \tau_{it}^{(r)} x_{ij}^2}, \quad (23)$$

for each i and j .

Note that (13) is concave in the alternative parameterization $\boldsymbol{\alpha}$ and $\tilde{\boldsymbol{\sigma}} = (\log \sigma_1^2, \dots, \log \sigma_g^2)^T$, and thus (21) and (22) globally maximize (13) over the parameter space $\Theta_\alpha \times \Theta_\sigma = \mathbb{R}^g \times (0, \infty)^g$. Furthermore, (14) is concave in $\boldsymbol{\beta}$, therefore (23) globally maximizes (14) over $\Theta_\beta = \mathbb{R}^{pg}$. The concavity of (13) and (14) can be deduced in a similar manner to the proof in Nguyen and McLachlan (2015, Thm. 2).

In summary, the block-wise MM algorithm for the maximization of (4) proceeds as follows. Initialize the algorithm with $\boldsymbol{\theta}^{(0)}$. Without loss of generality, at the $(r + 1)$ th step of the algorithm, perform one of the following updates:

- If $(r \bmod 3) + 1 = 1$, then perform the update (20) and set $\boldsymbol{\alpha}^{(r+1)} = \boldsymbol{\alpha}^{(r)}$, $\boldsymbol{\beta}^{(r+1)} = \boldsymbol{\beta}^{(r)}$, and $\boldsymbol{\sigma}^{(r+1)} = \boldsymbol{\sigma}^{(r)}$.
- If $(r \bmod 3) + 1 = 2$, then perform updates (21) and (22) for each i , and set $\boldsymbol{\pi}^{(r+1)} = \boldsymbol{\pi}^{(r)}$ and $\boldsymbol{\beta}^{(r+1)} = \boldsymbol{\beta}^{(r)}$.
- If $(r \bmod 3) + 1 = 3$, then perform the updates (23) for each i and j , and set $\boldsymbol{\pi}^{(r+1)} = \boldsymbol{\pi}^{(r)}$, $\boldsymbol{\alpha}^{(r+1)} = \boldsymbol{\alpha}^{(r)}$, and $\boldsymbol{\sigma}^{(r+1)} = \boldsymbol{\sigma}^{(r)}$.

2.5. Convergence analysis

The MM algorithm is iterated until some convergence criterion is met. In this article, we choose to use the absolute convergence criterion

$$\mathcal{F}_{n,\varepsilon}(\boldsymbol{\theta}^{(r+1)}) - \mathcal{F}_{n,\varepsilon}(\boldsymbol{\theta}^{(r)}) < \text{TOL},$$

where $\text{TOL} > 0$ is a small tolerance constant; see Lange (2013, Sec. 11.4) for details regarding the relative merits of convergence criteria. Upon convergence, we denote the final iterate of the algorithm to be the MPL estimate and write it as $\hat{\boldsymbol{\theta}}_n$.

Let $\boldsymbol{\theta}^{(\infty)}$ be a finite limit point of the MM algorithm, where $\boldsymbol{\theta}^{(\infty)} = \lim_{r \rightarrow \infty} \boldsymbol{\theta}^{(r)}$, or alternatively $\hat{\boldsymbol{\theta}}_n \rightarrow \boldsymbol{\theta}^{(\infty)}$ as $\text{TOL} \rightarrow 0$. Since each of the functions from Proposition 1 is concave over some bijective mapping of the parameters $\boldsymbol{\theta}$, and each of the updates (20)–(23) uniquely maximize their respective block-wise minorizers, the MM algorithm is a BSLM algorithm that adheres to the assumptions of Razaviyayn et al. (2013, Thm. 2); we can inspect these properties in a similar manner to the proofs of Nguyen and McLachlan (2015, Thms. 2 and 3). We thus have the following result.

Theorem 1. *If $\boldsymbol{\theta}^{(r)}$ is an iterate and $\boldsymbol{\theta}^{(\infty)}$ is a limit point of the block-wise MM algorithm for the MPL estimation of the L-MLR model (as given in Section 2.4) for some initialization $\boldsymbol{\theta}^{(0)}$, then the sequence $\mathcal{F}_{n,\varepsilon}(\boldsymbol{\theta}^{(r)})$ is monotonically increasing in r , $\boldsymbol{\theta}^{(\infty)}$ is a coordinate-wise minimizer of (4), and $\boldsymbol{\theta}^{(\infty)}$ is a stationary point of (4).*

Theorem 1 is a useful result considering that (4) is both multimodal and unbounded. Because of this fact, the MM algorithm should be run multiple times from different initializations $\boldsymbol{\theta}^{(0)}$ in order to obtain an appropriate limit point. The issues of multimodality and local roots of mixture model likelihoods are discussed in McLachlan and Peel (2000)[Sec. 2.12]. The explicit mitigation of the unboundedness of mixture log-likelihood functions has been addressed in Hathaway (1985) and Ingrassia and Rocci (2007, 2011). Further, a regression version of the approach of Chen et al. (2008) may also be devised to resolve the unboundedness problem.

In this article, we utilize the aforementioned approach of McLachlan and Peel (2000) in order to avoid local maxima. We did not require any explicit mitigation against the unboundedness of (4) as it did not arise in any of our simulations or applied computations.

3. Statistical inference

Let $\mathbf{x}_t$ be a realization of some random variable $\mathbf{X}_t$ with a well-behaved probability density $f(\mathbf{x}_t)$. Here, we leave the definition of well-behaved to be purposefully vague, as it is only required as a means of guaranteeing the uniform convergence of certain random functions. For argument sake, we can make $f(\mathbf{x}_t)$ well-behaved by supposing that it is a continuous density function defined over a compact domain. We will utilize well-behavedness in the same sentiment, albeit with different conditions with regard to Theorems 2 and 3. See the proofs of the theorems in Appendix A for details.

Given an appropriate density function $f(\mathbf{x}_t)$, we can write the joint density of any pair $(\mathbf{X}_t, Y_t)$ as $f(\mathbf{x}_t, y_t; \boldsymbol{\theta}) = f(\mathbf{x}_t) \sum_{i=1}^g \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \boldsymbol{\beta}_i, \sigma_i^2)$ and interpret Khalili and Chen (2007, Thm. 1) as follows.

Proposition 2. If $(\mathbf{X}_t, Y_t)$, for $t = 1, \dots, n$, is an IID random sample from a population with density function $f(\mathbf{x}_t, y_t; \boldsymbol{\theta}_0)$, where $f(\mathbf{x}_t)$ is well-behaved and $\boldsymbol{\theta}_0$ is the true value of the population parameter, then there exists a local maximizer $\hat{\boldsymbol{\theta}}_n$ of (2) for which

$$(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^T (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = O\left(n^{-1/2} \left[1 + \max_i \gamma_{in}\right]\right).$$

If we let $\gamma_{in} \rightarrow 0$ as $n \rightarrow \infty$ for each i , then Proposition 2 implies that the MPL estimators are root- n consistent.

Unfortunately, as mentioned by Khalili and Chen (2007, Thm. 1), we cannot select sequences γ_{in} such that $\hat{\boldsymbol{\theta}}_n$ is both sparse and root- n consistent simultaneously. That is, if we can decompose the population regression coefficients as $\boldsymbol{\beta}_{0i} = \{\boldsymbol{\beta}_{0i}^{[0]}, \boldsymbol{\beta}_{0i}^{[1]}\}$, for each i , where $\boldsymbol{\beta}_{0i}^{[1]}$ contains the non-zero coefficients and $\boldsymbol{\beta}_{0i}^{[0]} = \mathbf{0}$, then there is no choice of sequences γ_{in} that would both make $\hat{\boldsymbol{\theta}}_n$ root- n consistent and make $\mathbb{P}(\hat{\boldsymbol{\beta}}_n^{[0]} = \mathbf{0}) \rightarrow 1$ as $n \rightarrow \infty$. Here, we write $\boldsymbol{\beta}_0^{[k]} = (\boldsymbol{\beta}_{01}^{[k]T}, \dots, \boldsymbol{\beta}_{0g}^{[k]T})^T$ for $k = 0, 1$, and we write $\hat{\boldsymbol{\beta}}_n^{[k]}$ as the MPL estimator of $\boldsymbol{\beta}_0^{[k]}$.

Furthermore, no limit point of the MM algorithm for maximizing (4) can be sparse, due to the nature of the MM increments. Because of these facts, we propose the following two-step procedure for obtaining sparse estimates of $\boldsymbol{\beta}_0$ using the MPL estimator of the L-MLR.

3.1. Variable screening

Inspired by an idea of de Leeuw and Lange (2009), we can consider that a number is effectively zero if it falls below some machine-epsilon determined threshold. For example, in double precision, the machine determined threshold may be $M_\varepsilon = 10^{-16}$. Thus, for each i and j , we put coefficient β_{ij} into $\beta_i^{[0]}$ if $|\hat{\beta}_{ij}| \leq M_\varepsilon$. This variable thresholding constitutes the first step of our two-step procedure, and is analogous to the screening approach of Bühlmann and van de Geer (2011, Sec. 2.5).

In the second step of the procedure, we estimate the MLR model with all regression coefficients in $\beta_i^{[0]}$ set to zero. That is, we estimate the model

$$f(y_i | \mathbf{x}_t; \boldsymbol{\theta}^{[1]}) = \sum_{i=1}^g \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \tilde{\boldsymbol{\beta}}_i, \sigma_i^2), \quad (24)$$

where we can decompose the regression coefficients as $\tilde{\boldsymbol{\beta}}_i = \{\boldsymbol{\beta}_i^{[0]}, \boldsymbol{\beta}_i^{[1]}\}$ with $\boldsymbol{\beta}_i^{[0]} = \mathbf{0}$. Here, $\boldsymbol{\theta}^{[1]} = (\boldsymbol{\pi}^T, \boldsymbol{\alpha}^T, \boldsymbol{\beta}^{[1]T}, \boldsymbol{\sigma}^T)^T$ is the parameter vector of (24), where $\boldsymbol{\beta}^{[1]} = (\boldsymbol{\beta}_1^{[1]T}, \dots, \boldsymbol{\beta}_g^{[1]T})^T$.

Since we do not know in advance whether a coefficient is correctly set to zero or not, we must consider model (24) as misspecified and define the quasi-ML estimator of $\boldsymbol{\theta}^{[1]}$ as $\tilde{\boldsymbol{\theta}}^{[1]}$, where $\tilde{\boldsymbol{\theta}}^{[1]}$ is an appropriate local maximizer of the quasi-likelihood function

$$\mathcal{L}_n^{[1]}(\boldsymbol{\theta}^{[1]}) = \sum_{t=1}^n \log \sum_{i=1}^g \pi_i \phi(y_t; \alpha_i + \mathbf{x}_t^T \tilde{\boldsymbol{\beta}}_i, \sigma_i^2).$$

We can compute the quasi-ML estimate $\tilde{\boldsymbol{\theta}}_n^{[1]}$ using the block-wise MM algorithm from Section 2.5 by setting $\lambda_{in} = 0$ for each i . We note that this implies that the update (20) reduces to the simplified form of $\pi_i^{(r+1)} = \sum_{t=1}^n \tau_{it}^{(r)} / n$ for each i .

Since $\tilde{\boldsymbol{\theta}}_n^{[1]}$ is a sequence of quasi-ML estimators of an MLR, for $n \in \mathbb{N}$, the usual asymptotic theorems for such models apply. The following results can be proved via application of Theorems 4.1.2 and 4.1.3 of Amemiya (1985), respectively.

Theorem 2 (Consistency). Assume that $(\mathbf{X}_t, Y_t)$, for $t = 1, \dots, n$, is an IID sample from a well-behaved population, such $\boldsymbol{\theta}_0^{[1]}$ is defined as a local maximizer of $\mathbb{E} \log f(Y_t | \mathbf{X}_t; \boldsymbol{\theta}^{[1]})$, under the restriction that $\sigma_i^2 > \underline{\sigma}^2$, for $i = 1, \dots, g$ and some small $\underline{\sigma}^2 > 0$. If $\tilde{\boldsymbol{\theta}}_n$ is the set of roots of $\nabla \mathcal{L}_n^{[1]} = \mathbf{0}$ corresponding to local maxima of $\mathcal{L}_n^{[1]}$, for each $n \in \mathbb{N}$, and under the restriction $\sigma_i^2 > \underline{\sigma}^2$, then

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\inf_{\tilde{\boldsymbol{\theta}}^{[1]} \in \tilde{\boldsymbol{\theta}}_n} (\tilde{\boldsymbol{\theta}}^{[1]} - \boldsymbol{\theta}_0^{[1]})^T (\tilde{\boldsymbol{\theta}}^{[1]} - \boldsymbol{\theta}_0^{[1]}) > \varepsilon \right] = 0,$$

for every $\varepsilon > 0$. Here, we place an arbitrary valid element into $\tilde{\boldsymbol{\theta}}_n$ if there are no roots corresponding to maxima so that the set is never empty.

Theorem 3 (Asymptotic Normality). Carry all assumptions and definitions from theorem (Consistency). Let $\tilde{\boldsymbol{\theta}}_n^{[1]}$ be a sequence of quasi-ML estimators, obtained by choosing one element from each set $\tilde{\boldsymbol{\theta}}_n$, so that $\tilde{\boldsymbol{\theta}}_n^{[1]}$ converges to $\boldsymbol{\theta}_0^{[1]}$ (i.e. $\tilde{\boldsymbol{\theta}}_n^{[1]}$ is a sequence of

consistent roots). Assume that $\mathbf{I}_1(\boldsymbol{\theta}_0^{[1]})$ and $\mathbf{I}_2(\boldsymbol{\theta}_0^{[1]})$ are finite and nonsingular, where

$$\mathbf{I}_1(\boldsymbol{\theta}^{[1]}) = \mathbb{E} \left[\frac{\partial^2 \log f(Y_t | \mathbf{X}_t; \boldsymbol{\theta}^{[1]})}{\partial \boldsymbol{\theta}^{[1]} \partial \boldsymbol{\theta}^{[1]T}} \right]$$

and

$$\mathbf{I}_2(\boldsymbol{\theta}^{[1]}) = n^{-1} \mathbb{E} \left[\nabla \mathcal{L}_n^{[1]}(\boldsymbol{\theta}^{[1]}) \nabla \mathcal{L}_n^{[1]T}(\boldsymbol{\theta}^{[1]}) \right].$$

If $n^{-1} \partial^2 \mathcal{L}_n^{[1]} / \partial \boldsymbol{\theta}^{[1]} \partial \boldsymbol{\theta}^{[1]T} |_{\boldsymbol{\theta}_n^*}$ converges in probability to $\mathbf{I}(\boldsymbol{\theta}_0^{[1]})$, for every sequence $\boldsymbol{\theta}_n^*$ that converges in probability to $\boldsymbol{\theta}_0^{[1]}$, then $n^{1/2}(\tilde{\boldsymbol{\theta}}_n^{[1]} - \boldsymbol{\theta}_0^{[1]})$ is asymptotically normal with mean $\mathbf{0}$ and covariance matrix $\mathbf{I}_1^{-1}(\boldsymbol{\theta}_0^{[1]}) \mathbf{I}_2(\boldsymbol{\theta}_0^{[1]}) \mathbf{I}_1^{-1}(\boldsymbol{\theta}_0^{[1]})$.

The proof of both theorems can be found in [Appendix A](#) (Proofs). We note that the well-behaved population assumption in [Theorem 2](#) is further elaborated upon in its proof. Furthermore, the proof of [Theorem 3](#) provides some conditions for the validation of its assumptions.

Together, [Theorems 2](#) and [3](#) can be used to obtain confidence intervals for our L-MLR screened regression coefficients. This is a useful result as it can be seen as an alternative to the usual oracle theorems for regularized regressions that depend on more complex penalization functions than the lasso; see for instance [Khalili and Chen \(2007, Thm. 2\)](#).

3.2. Tuning parameter and components selection

In [Stadler et al. \(2010\)](#) an information-theoretic method using the BIC (Bayesian information criterion) of [Schwarz \(1978\)](#) is suggested for the choice of tuning parameters $\boldsymbol{\lambda} = (\lambda_{1n}, \dots, \lambda_{gn})^T$ and the number of components g in the MLR model. We shall follow the same approach and describe our process as follows.

Suppose that $(g_0, \boldsymbol{\lambda}_0) \in \{(g_1, \boldsymbol{\lambda}_1), \dots, (g_M, \boldsymbol{\lambda}_M)\}$ is a pair of parameters whereupon g_0 is the true number of components and $\boldsymbol{\lambda}_0$ results in the correct partitioning of regression coefficients, such that $\boldsymbol{\beta}_i^{[0]} = \boldsymbol{\beta}_{0i}^{[0]} = \mathbf{0}$. For each pair $(g_k, \boldsymbol{\lambda}_k)$, where $k = 1, \dots, M$, we perform the two-step procedure described in [Section 3.1](#) and obtain the parameter estimates $\tilde{\boldsymbol{\theta}}_k^{[1]} = (\boldsymbol{\pi}_k^T, \boldsymbol{\alpha}_k^T, \boldsymbol{\beta}_k^{[1]T}, \boldsymbol{\sigma}_k^T)^T$ for a screened MLR model. The BIC for the model can then be computed as

$$\text{BIC}_k = -2\mathcal{L}_n^{[1]}(\tilde{\boldsymbol{\theta}}_k^{[1]}) + (3g_k + p_k - 1) \log n, \quad (25)$$

where p_k is the dimensionality of $\boldsymbol{\beta}_k^{[1]}$ (i.e. the total number of non-zero regression coefficients in the model). The BIC rule for tuning parameters selection is to set $(g, \boldsymbol{\lambda}) = (\tilde{g}_k, \tilde{\boldsymbol{\lambda}}_k)$, where

$$\tilde{k} = \arg \min_k \text{BIC}_k.$$

We note that although the BIC has little theoretical support in the MLR setting, simulation studies such as those in [Grün and Leisch \(2007\)](#) and [Nguyen and McLachlan \(2016\)](#) indicate that the BIC tends to outperform or be comparable to other proposed criteria for such models.

4. Algorithm implementation

The proposed algorithm was implemented in the C++ programming language and a binary compiled for Mac OS X and source code are available at <https://github.com/lukelloydjones/LMLR>. The algorithm implementation is outlined in [Section 4.1](#). With regard to computing the $(r + 1)$ th iterate of $\boldsymbol{\pi}$ we use Brent's algorithm with reverse communication ([Brent, 2002](#)) to find the unique root of (19) in the interval (b^*, ∞) . The method requires the optimization over $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_g)^T$, which makes grid search prohibitively expensive for larger values of g . We use the ideas of [Wu and Lange \(2008\)](#), who suggest a shortcut to estimate $\boldsymbol{\lambda}$ by combining bracketing and the golden section search algorithm ([Kiefer, 1953](#)). We advance this approach by using golden section search to initialize the $\boldsymbol{\lambda}$ vector and then optimize over all parameters in $\boldsymbol{\lambda}$ via the Nelder–Mead method ([Nelder and Mead, 1965](#)). This procedure begins with an initial definition of the bounds of the golden section search algorithm. All elements of the $\boldsymbol{\lambda}$ vector are then set to the current update of the golden section search algorithm and a minimum of Eq. (25) is determined for the initial one dimensional section $(\lambda_{b1}, \lambda_{b2})$. The elements of the $\boldsymbol{\lambda}$ vector are then set to the best λ on $(\lambda_{b1}, \lambda_{b2})$, which initializes the starting simplex for the Nelder–Mead algorithm. The Nelder–Mead method then minimizes (25) over $\boldsymbol{\lambda}$. We propose that this formulation is a heuristic that is practically more efficient than grid search for the multi-dimensional optimization of $\boldsymbol{\lambda}$.

4.1. Algorithm summary

Algorithm 1

```

Initialize  $\lambda_{b1}$ ,  $\lambda_{b2}$ , TOL, maximum iterations,  $M_\epsilon$ ,  $\sigma$ ,  $\alpha$ ,  $\pi$  from parameter file
Read in csv format  $\mathbf{X}$  ( $n \times p$ ) and  $\mathbf{y}$  ( $n \times 1$ )
Golden section search algorithm start. Note: sets  $\lambda_i$  to be a common  $\lambda$  for all  $i \in \{1, \dots, g\}$ 
Set lower bound for golden search to  $\lambda_{b1}$ 
Set upper bound for golden search to  $\lambda_{b2}$ 
For initial  $\lambda$  bounds run BIC optimization function and return minimum BIC using Eqn. (25)
BIC optimization function start
  Read in from file starting values for all  $\beta_i$ 
  Pass initial  $\beta_i$  to MLR optimization function
  MLR optimization function start
    Initialise  $\lambda$ ,  $\beta_{0i}$ , TOL, maximum iterations,  $\epsilon$ ,  $\sigma$ ,  $\alpha$ ,  $\pi$  based on read parameters and current
    update of  $\lambda$ 
    Initialize objective function
    Set log-likelihood = 0.0
    for t := 1 to n do
      for i := 1 to g do
         $\mu_{ti} = \alpha_i + \mathbf{x}_t^T \beta_i$ 
        likelihood +=  $\pi_i \phi(y_t; \mu_{ti}, \sigma_k)$  od
        log-likelihood += log(likelihood) od
      Subtract lasso penalisation
      for i := 1 to g do
        log likelihood -=  $\pi_k \lambda_k \sum_t |\beta_k|$  od
    Initialize  $\tau_i$  and  $\omega_i$ 
    Begin main for loop
    for iteration := 1 to maximum iterations do
      Update  $\tau_i$ 
      for i := 1 to g do
         $\tau_i = \pi_k \phi(y_t; \mu_{tk}, \sigma_k) / f(y_i | \mathbf{x}_t; \theta)$  od
      Use Brent's algorithm with reverse communication to find unique root ( $\zeta^*$ ) of (19)
      Update  $\pi_i$  to by substituting  $\zeta^*$  into  $\pi_i^{(r+1)} = a_i / (\zeta^* + b_i)$ 
      Update  $\alpha_i$  by Eqn. (21)
      Update  $\sigma_i$  by Eqn. (22)
      Update  $\tau_i$  given new  $\alpha_i$  and  $\sigma_i$ 
      Update  $\beta_{ij}$  by Eqn. (23)
      Update log likelihood as above
      if | log-likelihood – log-likelihood old | < TOL return optimal  $\beta_i$  od else continue fi
    MLR optimization function end
  if  $|\beta_{ij}| < M_\epsilon$ , set  $\beta_{ij} = 0$ 
  Run MLR optimization function with thresholded  $\beta_i$  and  $\lambda = 0$ 
  Return BIC from unpenalized MLR optimization function
BIC optimization function end
Iterate golden section search algorithm over  $(\lambda_{b1}, \lambda_{b2})$  to find  $\lambda$  that minimizes (25)
if upper bound – lower bound < 0.1 return optimal  $\lambda$  and minimum BIC else continue fi
Golden section search algorithm end
Downhill simplex method start
  Initialize  $\lambda_i$  simplex with best  $\lambda$  from golden section search algorithm
  Downhill simplex method optimizes over BIC optimization function to find the best  $\lambda$  vector
Downhill simplex method end
Return best  $\lambda$  vector and BIC
Write  $\beta$ ,  $\lambda$ ,  $\sigma$ ,  $\alpha$ ,  $\tau$  and  $\pi$  to file

```

5. Analysis of simulated data

To evaluate the performance of the proposed algorithm we simulated data under the normal MLR model with two, three and four components. We based our simulation model on that presented in [Khalili and Chen \(2007\)](#) and extended their set of (n, p) scenarios using a similar structure to that presented in [Wu and Lange \(2008\)](#). Assuming that g is known, the model for the first simulation was a $g = 2$ model of form (1) where $\pi_1 = 0.5$, $\alpha_1 = -20$, $\alpha_2 = 20$, $\sigma_1 = 1$, and $\sigma_2 = 1$. For each repetition of the simulation, columns of $\mathbf{X}$ were drawn from a binomial $(2, q)$ distribution, with q sampled from a uniform distribution on $(0.05, 0.5)$. This corresponds to columns with entries 0, 1, or 2 with varying rates of success depending on the sampled q for that column. This resembles data from genetics, where the values 0, 1, and 2 represent the counts of a reference allele for a gene or single nucleotide polymorphism ([1000 Genomes Project Consortium et al., 2012](#)). The methodology is to be used in this context in future work and thus we simulated $\mathbf{X}$ in this way. The matrix $\mathbf{X}$ is column standardized to have mean 0 and variance 1 before multiplication with β_i . The vectors β_1 and β_2 have five non-zero elements each, with elements $1 \leq j \leq 5$ of β_1 and the elements $6 \leq j \leq 10$ of β_2 having an effect size of 5. Five scenarios were evaluated with varying values of n and p with 50 repetitions done for each.

The second simulation used model (1) but with a much more challenging set of initial parameters, which included $\pi_1 = 0.3$, $\alpha_1 = 0$, $\alpha_2 = 7$, $\sigma_1 = 1$, and $\sigma_2 = 1$, and the matrix $\mathbf{X}$ simulated as in the first simulation. The number of non-zero elements of β_i varied, with elements $1 \leq j \leq 3$ of β_1 equal to $(2, 1.5, 2)$ and elements $5 \leq j \leq 12$ of β_2 equal to $(0.5, -2, 1, 0.5, 0.5, 0.5, 2, -1)$. Again varying values of n and p were explored with 50 iterates performed for each of the n and p combinations.

The third simulation used model (1) but set $g = 3$. The value of n was fixed across scenarios but the values of p , α and π varied to challenge the methodology. The number of non-zero elements of β_i varied, with elements $1 \leq j \leq 2$ of β_1 equal to $(5, -5)$, $4 \leq j \leq 5$ of β_2 equal to $(-5, 5)$, and $p - 2 \leq j \leq p$ of $\beta_3 = (0, -5, 5)$ in the initial scenario and $\beta_3 = (2, 5, -5)$ in scenarios two and three. Across the three scenarios $p = (20, 100, 100)$ with $\alpha = (-20, 0, 20)$, $\alpha = (-10, 0, 10)$, and $\alpha = (-2, 0, 2)$. The final scenario was used to test the method's robustness to the populations being less well separated.

The fourth simulation used model (1) but set $g = 4$. The value of n was fixed but the values of p , and elements of α and π varied across scenarios to challenge the methodology. The number of non-zero elements of β_i varied, with elements $1 \leq j \leq 2$ of β_1 equal to $(5, -5)$, $4 \leq j \leq 5$ of β_2 equal to $(5, -5)$, $p - 4 \leq j \leq p - 3$ of $\beta_3 = (-5, 5)$ and $p - 2 \leq j \leq p$ of $\beta_4 = (0, -5, 5)$ in the first scenario and $\beta_3 = (2, 5, -5)$ in scenarios two and three. Across the three scenarios $p = (15, 50, 50)$ (less than simulation three as there are more groups) with $\alpha = (-30, -10, 10, 30)$, $\alpha = (-20, -10, 0, 20)$, and $\alpha = (-2, 0, 2, 4)$. The convergence of the means across scenarios was again used to test the method's robustness to the populations being less well separated. Post analysis of each of the simulation scenarios, for each replicate the mean squared error (MSE) between the predicted outcomes and the 'true' simulated outcomes from the L-MLR model was compared with those from multiple regression, or if $p > n$, marginal linear regression.

Table 1 reports the results from the first simulation based on 50 replicates and combinations of the sample size n and dimension p specified by $(n, p) = (200, 20), (200, 100), (500, 200), (200, 250)$, and $(500, 1000)$. For each of the scenarios β coefficients were estimated with high accuracy and precision. Reduced accuracy was seen when p became larger than n but precision remained approximately stable. The intercept terms α and mixing proportions π were well estimated with high precision and accuracy seen for both. As p increased the number of false non-zero elements (N_{NZ}) of β increased, with the mean for these parameters ranging from 0.04 to 0.19 across the scenarios, suggesting average small effects for these predictors. In each scenario the MSE from the L-MLR model was substantially smaller than that from marginal or multiple regression. Average computing time in seconds (on an Intel Core i7 CPU running at 2.8 GHz with 16 GB internal RAM) was recorded in the last column of Table 1 showing a large increase in computing time for the $(n, p) = (500, 1000)$ case.

Table 2 reports results from the more challenging second simulation, which again contained 50 replicates across $(n, p) = (200, 20), (500, 20), (200, 250)$ and $(350, 400)$. The optimization routine incorporated two λ coefficients, one for each of the regressor vectors that corresponded to the mixture components. For the $(n, p) = (500, 20)$ case, parameters were well estimated with low bias and high precision observed for the true non-zero elements of β . For the $(n, p) = (200, 20), (200, 250)$ and $(350, 400)$ cases, true non-zero elements of β were less well estimated with the true non-zero elements of β_1 and β_2 downwardly biased. For all simulation cases the intercept and mixing proportion parameters were well estimated with low bias and high precision seen for both. The number of false non-zero β elements increased with p with mean values being between 0.001 and 0.01 for all simulation cases respectively; again suggesting that although false non-zero elements are present the effect size is on average small. For the largest simulation scenario, computing time was in the order of half an hour with a standard deviation of approximately nine minutes.

Simulations three and four extended the two mixture simulations and provided more challenging scenarios in terms of distinct modes for each mixture component. Through experimentation we observed that an approximate minimum of $n = 1000$ was required before estimates of regression parameters began to be returned with high accuracy and precision, which drove our choice of $n = 1500$. For all scenarios in simulations three and four, the parameters were well estimated with low bias and high precision (Tables S1 and S2). The number of false non-zero β elements increased with p with mean values of falsely identified non-zero coefficients being of the order of 0.001. The precision of the parameter estimates decreased as the elements of α we made closer together. The difference between the MSE of the MLR model and multiple regression decreased as the elements of α were made closer together.

6. Analysis of real data

As a further test of the methodology, we investigated baseball salaries from the Journal of Statistics Education (www.amstat.org/publications/jse) with the intention to infer whether measures of batting performance affect an individual's salary. This was done to provide a baseline comparison with the foundational work of Khalili and Chen (2007) and for comparison with more recent data. These data contained salaries for 337 MLB players (batters only) from the year 1992, and 16 measures of batting performance from the year 1991. Each of the MLB players participated in at least one game in both the 1991 and 1992 seasons. As highlighted by Khalili and Chen (2007), the log(salary) histogram shows multi-modality making it a good candidate for a response variable under the MLR model (Fig. 1).

Each of the measures of batting performance is summarized in Table 3. The same set of interaction terms from Khalili and Chen (2007) was used in the analysis, which included further 16 interaction terms, making in total 32 predictors. Columns of $\mathbf{X}$ were standardized to have mean 0 and variance 1 for use with the methodology. For comparison, the stepAIC function from the MASS package (Venables and Ripley, 2002) of the R programming language (R Core Team, 2015) was used to implement model selection via the BIC for the standard linear model. For ease of comparison we reported the parameter estimates for β_1 and β_2 from the MIXLASSO method presented in Khalili and Chen (2007) with predictions from these values calculated assuming that $\mathbf{X}$ is column standardized (mean 0 and variance 1). Prediction with an unstandardized matrix yielded very poor results and thus we assumed that column-wise normalization was conducted, although it was not explicitly stated in Khalili and Chen (2007). We ran multiple analyses that included three sets of starting parameters for each of $g = (2, 3, 4)$. The starting value sets for β included: the estimates from Khalili and Chen (2007), with additional zero

Table 1

Summary of lasso-penalized mixture of linear regressions estimates and standard errors (below true values) from the first simulated scenario. Standard errors of estimates appear below parameter estimates. The number of false non-zero predictors (N_{NZ}) and the mean β estimate over all N_{NZ} are also reported. The mean squared error (MSE) for predictions from the L-MLR and multiple linear regression or, if $p > n$, marginal regression are reported for each of the simulations. Time is in seconds.

(n, p)	λ	$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{1,4}$	$\beta_{1,5}$	$\beta_{2,6}$	$\beta_{2,7}$	$\beta_{2,8}$	$\beta_{2,9}$	$\beta_{2,10}$	α_1	α_2	π_1	π_2	N_{NZ}	Mean NZ	MSE _{L-MLR}	MSE _R	Time
True values		5	5	5	5	5	5	5	5	5	5	−20	20	0.5	0.5					
(200, 20)	6.2	5.00	5.02	5.03	4.99	5.02	4.98	5.02	5.00	5.00	5.04	−20.0	20.0	0.499	0.501	10.5	0.10	1.2	414.9	1.4
	2.1	0.12	0.10	0.11	0.10	0.09	0.10	0.11	0.12	0.13	0.11	0.10	0.10	0.002	0.002	2.6	0.02	0.14	15.8	0.3
(200, 100)	36.1	5.00	4.97	4.97	4.99	4.97	4.97	4.99	4.99	4.97	4.98	−20.0	20.0	0.500	0.500	4.5	0.19	1.1	230.1	12.7
	2.1	0.12	0.11	0.11	0.08	0.11	0.12	0.13	0.12	0.11	0.09	0.09	0.09	0.003	0.003	3.24	0.06	0.14	24.7	2.9
(500, 200)	48.6	4.98	5.00	4.98	4.98	4.97	4.99	4.98	4.99	4.98	4.98	−20.0	20.0	0.500	0.500	29.4	0.11	1.2	279.4	106
	1.0	0.07	0.08	0.06	0.07	0.06	0.06	0.07	0.08	0.06	0.08	0.05	0.06	0.002	0.002	8.2	0.01	0.10	16.9	20
(200, 250)	40.9	4.97	4.98	4.96	4.99	4.98	4.95	4.97	4.98	4.97	4.97	−19.9	20.0	0.500	0.500	19.7	0.16	1.3	820.5	119.5
	3.2	0.11	0.12	0.11	0.11	0.12	0.15	0.13	0.13	0.13	0.13	0.12	0.11	0.002	0.002	48.7	0.12	0.61	130.5	99.5
(500, 1000)	37.1	4.85	5.05	4.80	5.04	4.81	5.05	4.83	5.07	4.83	5.05	−20.0	20.0	0.500	0.500	564.7	0.04	28.0	1793.4	39,918
	2.1	0.11	0.10	0.08	0.11	0.09	0.13	0.11	0.10	0.10	0.11	0.09	0.09	0.002	0.002	17.1	0.002	10.6	196.3	27,454

Table 2

Summary of lasso-penalized mixture of linear regressions estimates and standard errors from the second simulated scenario. Standard errors of estimates appear below parameter estimates. The number of falsely identified non-zero predictors (N_{NZ}) and the mean β estimate over all N_{NZ} are reported. The mean squared error (MSE) for predictions from the L-MLR and multiple linear regression or, if $p > n$, marginal regression are reported for each of the simulations. Time is in seconds.

(n, p)	λ_1	λ_2	$\beta_{1,1}$	$\beta_{1,2}$	$\beta_{1,3}$	$\beta_{2,5}$	$\beta_{2,6}$	$\beta_{2,7}$	$\beta_{2,8}$	$\beta_{2,9}$	$\beta_{2,10}$	$\beta_{2,11}$	$\beta_{2,12}$	α_1	α_2	π_1	π_2	N_{NZ}	Mean NZ	MSE _{L-MLR}	MSE _R	Time
True values			2	1.5	2	0.5	−2	1	0.5	0.5	0.5	2	−1	0	7	0.3	0.7					
(200, 20)	38	32	1.87	1.35	1.81	0.45	−1.83	0.91	0.47	0.44	0.45	1.83	−0.89	−0.02	7.0	0.29	0.71	3	0.00	1.73	14.1	14
	13	13	0.51	0.48	0.56	0.17	0.55	0.32	0.18	0.19	0.17	0.47	0.31	0.46	0.23	0.03	0.03	5	0.13	2.6	1.1	5
(500, 20)	14	15	1.99	1.49	2.01	0.50	−2.00	1.00	0.50	0.50	0.49	2.00	−1.00	0.01	7.0	0.30	0.70	17	0.001	0.98	14.8	16
	3	4	0.11	0.10	0.08	0.06	0.06	0.06	0.05	0.05	0.06	0.06	0.05	0.08	0.06	0.01	0.01	3	0.02	0.15	2.2	3
(200, 250)	49	48	1.87	1.26	1.74	0.37	−1.87	0.90	0.41	0.42	0.36	1.93	−0.88	−0.09	7.0	0.31	0.70	46	−0.01	1.89	61.4	789
	67	89	0.19	0.36	0.40	0.22	0.41	0.29	0.22	0.22	0.21	0.13	0.28	0.24	0.14	0.02	0.02	30	0.06	0.87	7.4	287
(350, 400)	60	59	1.87	1.40	1.88	0.47	−1.98	0.99	0.47	0.47	0.47	1.96	−0.96	−0.04	7.0	0.31	0.69	98	−0.01	1.51	56	2193
	6	5	0.10	0.12	0.11	0.10	0.08	0.09	0.10	0.10	0.09	0.07	0.07	0.14	0.06	0.01	0.01	42	0.04	0.28	5.1	524

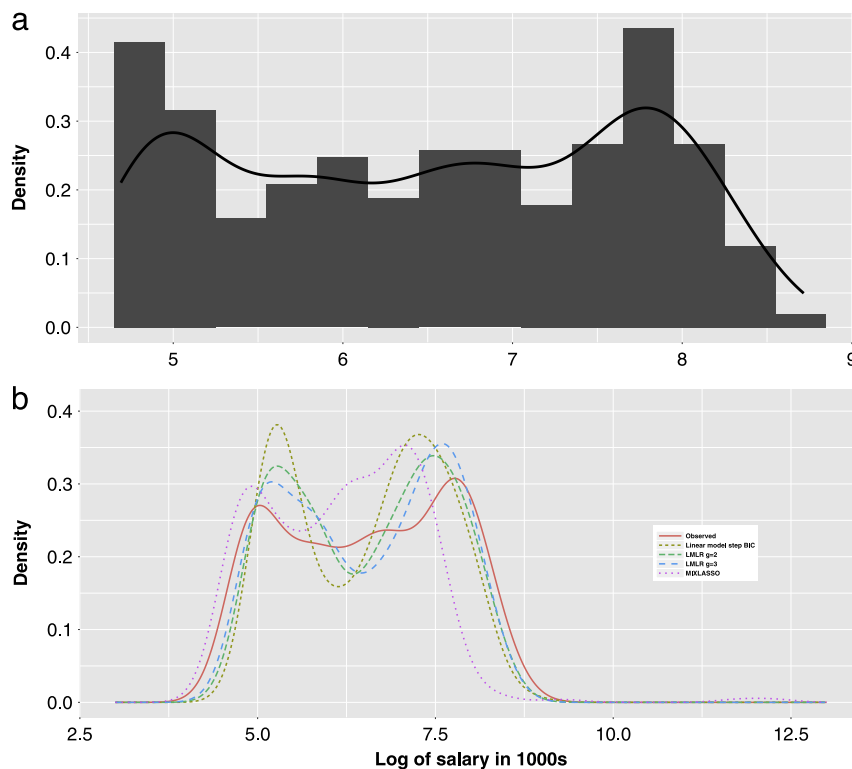


Fig. 1. Summary of raw and predicted $\log(\text{salary})$ for batters from Major League Baseball in years 1991/92. Panel (a) depicts a histogram of the log of salaries in 1000s for all 337 batters with the fitted density line. Panel (b) shows the predicted densities of log salaries in 1000s from the stepwise-BIC, MIXLASSO, and L-MLR ($g = (2, 3)$) models from Table 3 with the observed data density also plotted for reference.

vectors for the $g = 3$ and 4 cases; duplicated marginal regression coefficients screened via p -value threshold of 0.05/32; and a set of vectors with elements drawn from the $N(0, 0.1)$ distribution. Mixing proportions and intercept starting values were varied around those of Khalili and Chen (2007) for each run.

The $g = 3$ model with the starting values from Khalili and Chen (2007) had the lowest BIC (573.6) across all analyses, with the $g = 2$ model being ranked second (BIC=580.1). The predicted logged salaries from the L-MLR $g = 3$ model had a MSE of 0.07 and a regression (Y_{pred} on Y_{obs}) R^2 of 0.95. The predicted logged salaries from the L-MLR $g = 2$ model had a MSE of 0.10 and a regression (Y_{pred} on Y_{obs}) R^2 of 0.93. The predicted logged salaries from the stepwise-BIC linear model showed a MSE of 0.27 and R^2 of 0.80. The predicted logged salaries from MIXLASSO had a MSE 0.58 and R^2 of 0.67. These results suggest that the L-MLR model had the smallest MSE and explained the largest proportion of variance for the baseball salary data from the 1991/92 MLB seasons. It is noted that the MIXLASSO algorithm is similar to that of Stadler et al. (2010), where approximate updates are used for the mixture model mixing proportions in the maximization-step of their algorithm. Therefore, the improvement in model prediction may be attributed to the exact updates for the mixing proportions used in the L-MLR analysis and the more complete tuning of the lasso penalty parameters.

A plot of the fitted posterior probabilities of membership of the first and second components of the mixture (τ_1 and τ_2) on the observed logged salaries for the $g = 3$ model revealed that the L-MLR model partitioned the batters into two primary groups (Figure S1). These groups coincided with a group of batters that were low and highly paid, and a group of averaged paid batters (Figure S2). A third group of four individuals with low salaries was identified; their predicted values were dependent on their arbitration eligibility (Table 3). For the low and high paid set of batters, the predictors that lead to variation included hits, runs batted in, free agent eligibility and agent eligibility. In contrast, the component two batters appeared to have average salaries that were differentiated by hits and interactions between runs batted in and agency eligibility. The predicted values from the stepwise-BIC linear model showed multi-modality and was a better predictor than the MIXLASSO model based on MSE and prediction R^2 . Free agent eligibility, arbitration eligibility, hits and runs batted in were also positive predictors for this model. Strikeouts and free agency were negative predictors. The predictors for this model were generally in line with those from L-MLR. However, the inference as to whether these predictors were associated with any latent sub-groups could not be made. The results presented in Khalili and Chen (2007) did not include a presentation of how the individuals were partitioned across the two subgroups. However, we reconstructed their component densities and observed that their method partitions players into lower and higher paid groups (Figure S2). Overall, the conclusions

Table 3

Summary of stepwise-BIC (SW-BIC), MIXLASSO (Khalili and Chen, 2007), and lasso-penalized mixture of linear regressions (L-MLR) estimates from each mixture component (C_1 – C_3) for baseball salary data from the 1990s. Covariate acronyms are AVG—Batting Average, OBP—On Base Percentage, R—Runs, H—Hits, 2B—Doubles, 3B—Triples, HR—Home Runs, RBI—Runs Batted In, BB—Walks, SO—Strikeouts, SB—Stolen Bases, ERS—Errors, FAE—Free Agency Eligibility, FA—Free Agent in 1991/92, AE—Arbitration Eligibility, ARB—Arbitration in 1991/92.

Covariates	SW-BIC	MIXLASSO		L-MLR		L-MLR		
		C_1	C_2	C_1	C_2	C_1	C_2	C_3
Intercept	6.54	6.41	7.00	6.46	6.59	5.17	6.58	6.60
π	–	0.72	0.28	0.24	0.76	0.02	0.32	0.66
AVG	–	–	–0.32	–	–0.05	–	–	–
OBP	–	–	0.29	–	–	–	–	–
R	–	–	–0.70	–	0.02	–	–	–
H	0.26	0.20	0.96	0.19	0.18	–	0.28	0.16
2B	–	–	–	0.28	–	–	–	–
3B	–	–	–	–	–	–	–	–
HR	–	–0.19	–	–	–	–	–	–
RBI	0.30	0.26	–	0.25	0.20	–	–	0.11
BB	–	–	–	–	0.09	–	0.02	0.06
SO	–0.14	–	–	–	–0.12	–	–	–
SB	–	–	–	–	0.06	–	–	0.05
ERS	–	–	–	–	–	–	–	–
FAE	0.83	0.79	0.70	–	0.94	–	0.01	1.03
FA	–0.18	0.72	–	–	–	–	–	–
AE	0.53	0.15	0.50	–	0.69	0.76	–	0.72
ARB	–	–	–0.36	–	–	–	–	–
AVG×FAE	–	–0.21	–	–	–	–	–	–
AVG×FA	–	0.63	–	–	–	–	–	–
AVG×AE	–	0.34	–	–	–0.11	–	–	–0.13
AVG×ARB	–0.19	–	–	–	–	–	–	–
R×FAE	–	–	–	–	–	–	0.18	–
R×FA	–	0.14	–0.38	–	–	–	–	–
R×AE	–	–	–	–	0.04	–	–	0.07
R×ARB	–	–0.18	0.74	–	–	–	–	–
HR×FAE	–	–	–	–	0.10	–	–	0.07
HR×FA	0.11	–	–	–	0.02	–	–	0.04
HR×AE	–	–	0.34	–	–	–	–	–
HR×ARB	–	–	–	–	0.04	–	–	–
RBI×FAE	–	0.29	–0.46	–	–	–	0.31	–
RBI×FA	–	–0.14	–	–	–	–	–	–
RBI×AE	–	–	–	–	0.05	–	0.37	0.08
RBI×ARB	0.19	–	–	–	–	–	–	–

appeared similar to Khalili and Chen (2007) with the different forms of agency eligibility having the largest effects. We agree with the remark of Khalili and Chen (2007) that from these data a player's worth is not always reflected in his salary.

To update these analyses, we gathered data from modern MLB batters from the 2011–15 seasons. Salaries for individuals were acquired from the website <http://www.usatoday.com/sports/mlb/salaries/> for all players in the MLB over these seasons. Salary values are based on documents obtained from club officials from the MLB Players Association. Incentive clauses and deferred payments are not included in the value. Salary values do not include money paid or received in trades or for players who have been released. Measures of batting performance for these players were gathered from the website <http://espn.go.com/mlb/statistics>, which include standard, expanded, and 'sabermetric' batting statistics for all batters in the MLB over the 2011–15 seasons. These seasons were played under the new bargaining agreement established in 2011 between the 30 Major League Clubs and the Major League Baseball Players Association (Major League Baseball Players Association, 2011). For each season, statistics were matched to players salaries in the R programming language with a total of 37 predictors available for each batter. The logged salaries (in 1000s) were again used as a response variable for the analysis. The distribution of combined modern salaries had three modes, which included a spike around \$600,000 suggesting a large number of MLB players on a base salary, and two broader density components (Figure S3). These three modes are likely to reflect the salary structure across Major League Baseball, which is dependent on the 2011 agreement between the 30 Major League Clubs and the Major League Baseball Players Association. In Section 6 of this agreement the rules stipulate that the minimum salary for a player per season, in the MLB, shall be \$480,000 to \$520,000 for the 2012 to 2016 seasons. Any player with a total of three or more years of Major League service, but with less than six years of service, may submit the issue of the player's salary to final and binding arbitration without the consent of the club (Major League Baseball Players Association, 2011). Arbitration allows players to enter into salary negotiations based on their performance, with the effect of eligibility for salary arbitration on player salaries often being dramatic (Abrams, 2010). At the salary arbitration stage the player's salary is set in comparison with other players around the league with similar performance statistics and major league experience (Abrams, 2010). Players with six or more years of Major League service who have not executed a contract for the next succeeding season are eligible to become a free agent. The progression to arbitration eligibility or free agent status is

associated with a large increase in the capacity of a player to earn a much larger salary. This induces a correlation between time played in the major league and pay, which is well known in the economics literature surrounding baseball (Scully, 1974; Hakes and Sauer, 2006). It is not until players reach free agent status that they may negotiate contracts with any team in a truly competitive labor market, often leading to a further dramatic increase in salary (over arbitration status) (Brown et al., 2015). However, players of rare talent earn salaries apparently in excess of their relative contribution to the team, which introduces a potential non-linearity into the performance salary model (Scully, 1974).

When modeled, no predictors were seen for the spike for minimum wage players as there was little variation around this value. We therefore chose to eliminate this spike for each year analyzed with a threshold set depending on the spike for that year (approximately \$665,000 to \$1,800,000 (USD) depending on the season). This left data sets comprising $n = 110, 94, 100, 112$, and 99 for the 2011–15 seasons respectively. Seasons were analyzed separately using the L-MLR model and the stepwise-BIC linear model. As contracts for batters in the MLB often extend for many years (5–10 years) a combined analysis over the five year period was also performed. This was done by averaging statistics and salaries for players who were present in the MLB over all or some of the years. This resulted in a data set of 215 batters (excluding those less than 6.5 on the logged salary scale) from the 2011–15 seasons. The primary motivation for this last analysis was the long term contracts present in the MLB and to increase sample size. For each year and the combined set, we ran multiple analyses that included three sets of starting parameters for each of $g = (2, 3, 4)$ models. The starting value sets for β included: the estimates from the stepwise-BIC duplicated for each component; duplicated marginal regression coefficients screened via p -value threshold of 0.05/37; and a set of vectors with elements drawn from the $N(0, 0.1)$ distribution.

For each of the analyses the $g = 2$ model with the marginal regression starting coefficients had the lowest BIC. Therefore, we only report the results from these analyses. Years 2011–2015 showed MSEs of (0.05, 0.06, 0.06, 0.14, 0.06) for L-MLR $g = 2$ model and (0.21, 0.19, 0.35, 0.50, 0.27) for the stepwise-BIC model, with similar patterns in R^2 seen (0.88, 0.86, 0.88, 0.80, 0.88) across the seasons. Predicted densities showed a good characterization of the multi-modality in the logged salaries for the L-MLR model, with the stepwise-BIC not being able to model this (Fig. 2). Plotting the observed salaries versus τ_1 for each of the years modeled, showed that the method partitions the salaries of modern batters into lower and higher salaries, which is in contrast to the salaries observed in the 1990s (Figure S1). Thus, the components of β contain predictors that explain variation for the lower paid batters and higher paid batters. Across seasons, predictors were not consistent for both the stepwise-BIC and L-MLR models.

The averaged analysis showed a MSE of 0.12 and R^2 of 0.84 for the L-MLR method, with the stepwise-BIC showing a MSE of 0.49 and R^2 of 0.32. Positive predictors for the highest paid batters (mean salary of ≈ 7.3 million dollars) included runs, runs batted in, total plate appearances, and intentional walks (Table 4). Negative predictors for the highest paid batters included doubles, strikeouts, wins against replacement, games played, and sacrifice hits. For medium paid batters (mean salary of ≈ 1.9 million dollars) no strong predictors were observed.

7. Discussion

In this work we have developed and implemented an algorithm that performs simultaneous model selection and estimation for a mixture of linear regressions model via lasso-penalized regression. The algorithm allows for coordinate-wise updates of parameters, and generates globally-convergent sequences of estimates that monotonically increase the penalized log-likelihood function, via the MM algorithm paradigm. We overcame the difficult step of correctly updating the mixing proportions for the mixture distribution by showing that there must exist a unique root of the FOC for the Lagrangian. Fast root-finding algorithms can then be exploited to find this root, which is used to update the mixing proportions in each iteration of the algorithm. This process is efficient and facilitates an extensible algorithm that can easily fit models with more than two components. We implemented the algorithm in a C++ program that is practically useful and is computationally efficient, allowing for large problems to be solved in reasonable time. We explored the use of a two-step procedure for optimizing the λ vector that uses the golden section search algorithm to find an initial optimum on a predefined one-dimensional section, where λ is fixed to be the same value for each element of λ . The nonlinear optimization method of Nelder and Mead is then used to minimize the BIC criterion over the λ vector.

Several factors make penalized regression an ideal exploratory data analysis tool. One is the avoidance of the issues of multiple testing (Wu and Lange, 2008). Another is the generation of a predictive model that can be used to interpolate or extrapolate from the data. However, the generation of this predictive model requires (in our method) the minimization of the BIC over λ , which we perform via numerical methods. The Nelder–Mead technique is a heuristic search method that can converge to non-stationary points and thus there is the potential that λ may not be a minimum of the BIC objective function. This is a practical trade-off between the computational demand of exhaustive grid search, which is prohibitive when g is large, or when the grid is fine, and the potential location of a non-stationary point. We also assume that the location of a shared λ via golden section search, is a good starting value for initializing the simplex of the Nelder–Mead method. Although the properties of Nelder–Mead are not known, to the best of our knowledge, we are the first to consider any sophisticated optimization over the λ vector in a mixture scenario. Although lasso-constrained l_2 -regression is consistent, parameter estimates are biased toward zero in small samples (Wu and Lange, 2008). For this reason, once we have identified the active parameters for a given value of the tuning vector λ , we re-fit the model without penalty. As noted by Wu and Lange (2008), if one uses the lasso-estimated parameters instead of the refitted parameters, then there will be a bias toward

Table 4

Summary of stepwise-BIC (SW-BIC) and lasso-penalized mixture of linear regressions for components one and two (C_1 and C_2) estimates (marginal regression starting values) from MLB data from the 2011–2015 seasons. Covariate acronyms are AB—At Bats, R—Runs, H—Hits, 2B—Doubles, 3B—Triples, HR—Home Runs, RBI—Runs Batted In, SB—Stolen Bases, CS—Caught Stealing, BB—Walks, SO—Strikeouts, AVG—Batting Average, OBP—On Base Percentage, SLG—Slugging Percentage, OPS—OPS = OBP + SLG, WAR—Wins Above Replacement, GP—Games Played, TPA—Total Plate Appearances, PIT—Number of Pitches, P/PA—Pitches Per Plate Appearance, TB—Total Bases, XBH—Extra Base Hits, HBP—Hit By Pitch, IBB—Intentional Walks, GDP—Grounded Into Double Plays, SH—Sacrifice Hits, SF—Sacrifice Flies, RC—Runs Created, RC27—Runs Created Per 27 Outs, ISP—Isolated Power, SECA—Secondary Average, GB—Ground Balls, FB—Fly Balls, G/F—Ground Ball to Fly Ball Ratio, AB/HR—At Bats Per Home Run, BB/PA—Walks Per Plate Appearance, BB/K—Walk to Strikeout Ratio.

Covariates	Season 2011			Season 2012			Season 2013			Season 2014			Season 2015			Combined		
	SW-BIC	C_1	C_2	SW-BIC	C_1	C_2	SW-BIC	C_1	C_2	SW-BIC	C_1	C_2	SW-BIC	C_1	C_2	SW-BIC	C_1	C_2
Intercept	–	8.52	9.13	–	8.41	9.29	–	8.10	9.27	–	7.68	9.20	–	8.29	9.42	–	7.54	8.90
π	–	0.39	0.61	–	0.42	0.58	–	0.35	0.65	–	0.25	0.75	–	0.35	0.65	–	0.26	0.74
AB	–	–	–	–10.8	–	–	–	–	–	–	–	–	–12.4	–	–	–	–	–
R	–	–	0.05	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0.19
H	–	–	–	28.5	–	–	0.50	–	–	–3.1	–	–	25.1	–	–	0.28	–	–
2B	–	–	–	3.7	0.02	–	–	–	–	–	–	–	2.9	–	–	–	–	–0.16
3B	–	–	0.06	2.7	–	–0.08	–	–	–	–	–	–	1.6	–	–0.19	0.21	–	–
HR	–	–0.04	–	14.2	–	–	0.57	–	–	–	–	–	15.0	–	–	0.80	–	–
RBI	–	0.06	0.28	–	–	–	–	–	0.05	–	–	0.14	–	0.19	–	0.41	–	0.22
SB	–	–	–	2.9	–	–	–	0.09	–	–	–	–	1.8	–0.01	–	–	–	–
CS	–0.13	–	–	–1.8	–	–0.02	–	–	–	–	–	–	–1.6	–	–0.07	–	–0.01	–
BB	–	–	–	9.4	–	–	0.36	–	–	–2.5	–	–	12.2	–	–	–	–	–
SO	–	–	–	–0.60	–	–	–0.18	–	–	–	–	–	–	–	–	–0.17	–	–0.14
AVG	–	–	–	–	–	–	–	–0.14	0.05	–	–	–	–	–	–	–	–	–
OBP	–	–	–	–	0.12	0.13	–	–	0.10	2.2	–	–	–8.1	–	–	–	–	–
SLG	–	–	–	–1.6	–	–	–	–	–	–	–	–	–	–	–	–	0.08	–
OPS	–	–	–	–	–	–	–0.58	–	–	–	–	–	13.0	–	–	–	–	–
WAR	–0.25	–	–0.29	–	–	–	–	–	–	–	–0.04	–	–	–0.25	–	–0.33	–	–0.21
GP	–0.18	–0.29	–0.20	–	–	–	–0.48	–	–	–0.40	–0.04	–	–	–	–	–0.31	–	–0.34
TPA	–	–	–	–	–	–	–	–	–	2.9	–	–	–	–	–	–	–	0.49
PIT	–	–	–	–	–	–	–	0.03	–	–	–0.16	–	3.5	–	–	–	–	–
P/PA	–	–	0.06	–	–0.15	–	–	–	–	–	–	–	–1.7	–	–	–	–	–
TB	–	0.48	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
XBH	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
HBP	–	–	–	2.3	0.07	–	–	0.09	–	–0.41	–	–	2.7	–	–	–	–0.09	–
IBB	–	0.13	–	0.15	–	–	–	–	–	0.26	–	–	–	–	–	–	–	0.11
GDP	–	0.22	–	–3.2	0.15	0.14	–	0.30	–	–	–0.18	–	–3.9	0.08	0.08	–	–	–
SH	–0.23	–	–0.39	–	–	–	–	–	–0.10	–	–	–0.14	–	–0.05	–	–	–0.07	–0.13
SF	–	0.11	–0.13	0.29	–	0.14	–	–	–	–	–	–	–	–	–	–	–	–
RC	–	–0.05	–	–30.0	0.03	–	–	–	–	–	–	–	–35.8	–	–	–	–	–
RC27	–	–	–	–	–	–	–	–	–	–	–	–	–2.9	–	–	–	–	–
ISOP	–	–	0.09	–	–	–	–	0.06	–	–	–	–	–10.8	–	–	–1.1	–	–
SECA	0.46	–	–	2.0	–	–	–	0.17	–	–	–	–	7.0	–	–	0.55	–	–
GB	0.44	–	0.35	–	–	–	–	0.14	–	–	–	–	1.5	–	–	–	–	–
FB	0.20	–	–	–	0.14	–	–	–	–	–	–	–	–	–	0.07	–	–	–
G/F	–	–	–0.09	–	–	–	–	–	–0.13	–	–	–	–1.2	–	–0.05	–	–	–
AB/HR	–	–	–	–	–0.01	–	–	–	–	–0.24	–	–	–	–	–0.06	–0.17	–	–
BB/PA	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
BB/K	–	0.10	–	–	–	–	–	0.13	–	–	–	–	–	0.14	–	–	–	–

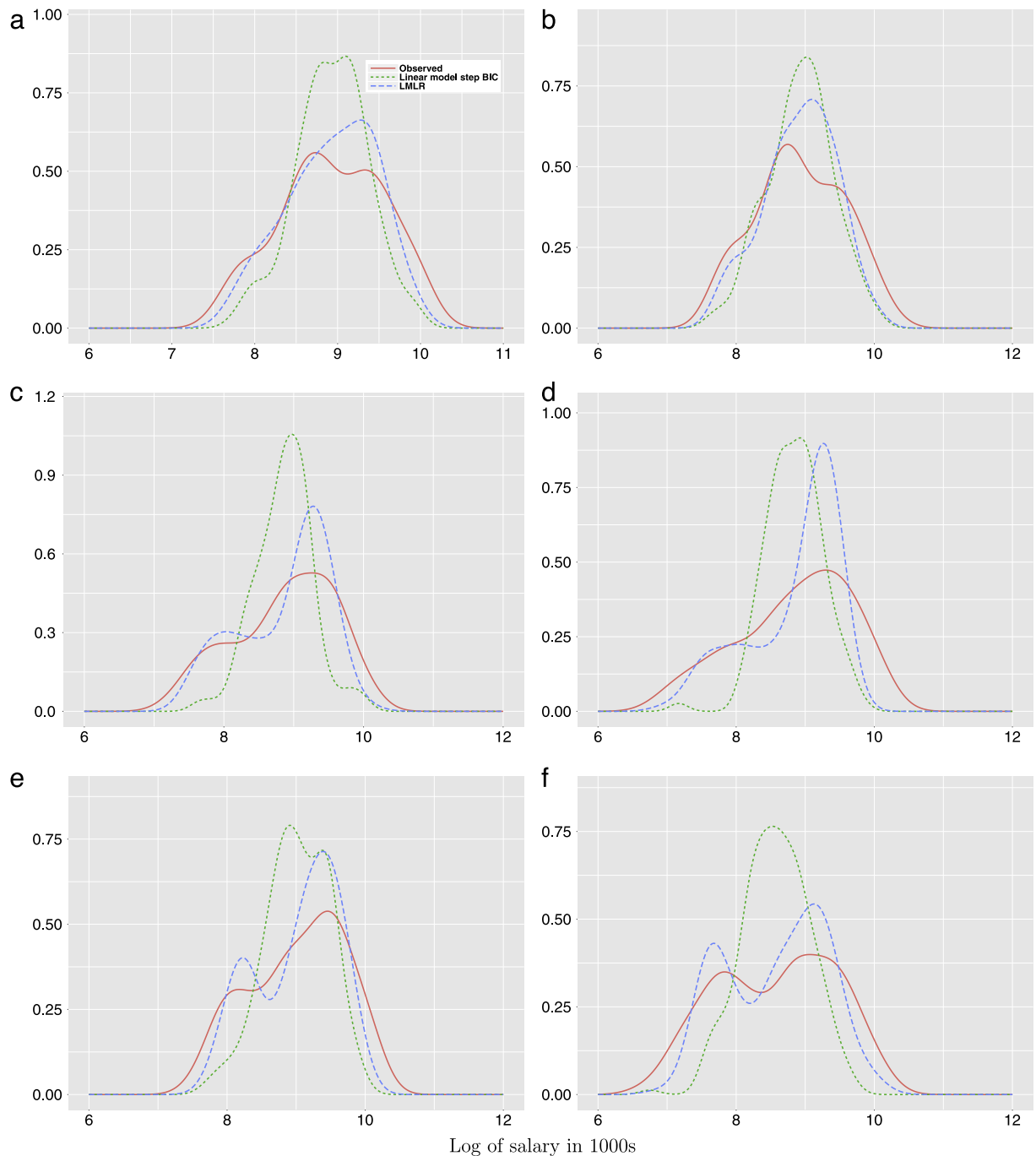


Fig. 2. Summary of densities for predicted and observed logged salaries (in 1000s of dollars) of batters from the MLB from seasons 2011–15. Panels (a) to (e) depict densities of logged salaries for the 2011–15 seasons with panel (f) depicting the predicted and observed densities of averaged logged salaries over all seasons. In each panel the densities of the observed logged salaries, the predicted values from the L-MLR model, and the predicted values from the stepwise-BIC linear model are plotted.

the inclusion of irrelevant covariates in the preferred model under any selection scheme that is a function of the estimated parameter vector (e.g., the BIC).

In future work, the computational efficiency of the algorithm will be investigated with possible significant improvements to be expected with the implementation of a coordinate-descent algorithm (Wu and Lange, 2008), especially for scenarios where $p \gg n$. Like our MM algorithm, coordinate descent algorithms allow for an avoidance of large matrix operations.

Furthermore, they also permit the ability to test for whether costly operations for a predictor are required, which reduces updates to only those contributing predictors for a given λ . This is in contrast to the current algorithm, which requires all predictors to be updated in each iteration. We reiterate that the source code for the presented algorithm is publicly available <https://github.com/lukelloydjones/LMLR>. It is our intention, that with a more computationally efficient algorithm and implementation, to deploy our methodology on heterogeneous population problems in genomics. Our simulations and applications demonstrate that the modeling of heterogeneity in regression data is important for accurate prediction, as demonstrated by comparisons against stepwise-BIC. The solution to the mixing proportion updates is not only applicable in the L-MLR context, but also in other lasso-constrained mixture of regressions models, such as the logistic regressions model (also suggested in [Khalili and Chen, 2007](#)) or for Laplace mixtures ([Nguyen and McLachlan, 2016](#)) with lasso constraints.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.csda.2017.09.003>.

References

- 1000 Genomes Project Consortium, et al., 2012. An integrated map of genetic variation from 1092 human genomes. *Nature* 491, 56–65.
- Abrams, R., 2010. *The Money Pitch: Baseball Free Agency and Salary Arbitration*. Temple University Press, Philadelphia.
- Amemiya, T., 1985. *Advanced Econometrics*. Harvard University Press, Cambridge.
- Breiman, L., et al., 2001. Statistical modeling: The two cultures (with comments and a rejoinder by the author). *Statist. Sci.* 16 (3), 199–231.
- Brent, R.P., 2002. *Algorithms for Minimization Without Derivatives*. Dover Publications, Mineola, New York.
- Brown, D.T., Link, C.R., Rubin, S.L., 2015. Moneyball after 10 years: How have major league baseball salaries adjusted? *J. Sports Econ.* 1–16.
- Bühlmann, P., van de Geer, S., 2011. *Statistics for High-Dimensional Data*. Springer, New York.
- Chen, J., Tan, X., Zhang, R., 2008. Inference for normal mixtures in mean and variance. *Statist. Sinica* 443–465.
- Compiani, G., Kitamura, Y., 2016. Using mixtures in econometric models: a brief review and some new results. *Econom. J.* 19 (3).
- de Leeuw, J., Lange, K., 2009. Sharp quadratic majorization in one dimension. *Comput. Statist. Data Anal.* 53 (7), 2471–2484.
- Dempster, A.P., Laird, N.M., Rubin, D.B., 1977. Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 39, 1–38.
- DeSarbo, W.S., Cron, W.L., 1988. A maximum likelihood methodology for clusterwise linear regressions. *J. Classification* 5, 249–282.
- De Veaux, R.D., 1989. Mixtures of linear regressions. *Comput. Statist. Data Anal.* 8, 227–245.
- Fullerton Jr., T.M., Peach, J.T., 2016. Major League Baseball 2015, what a difference a year makes. *Appl. Econ. Lett.* 1–5.
- George, E.I., 2000. The variable selection problem. *J. Amer. Statist. Assoc.* 95, 1304–1308.
- Greene, W.H., 2003. *Econometric Analysis*. Prentice Hall.
- Grün, B., Leisch, F., 2007. Fitting finite mixtures of generalized linear regressions in R. *Comput. Statist. Data Anal.* 51, 5247–5252.
- Hakes, J.K., Sauer, R.D., 2006. An economic evaluation of the Moneyball hypothesis. *J. Econ. Perspect.* 20, 173–185.
- Hastie, T., Tibshirani, R., Friedman, J., 2009. *The Elements Of Statistical Learning*. Springer, New York.
- Hathaway, R.J., 1985. A constrained formulation of maximum-likelihood estimation for normal mixture distributions. *Ann. Statist.* 13, 795–800.
- Hennig, C., 2000. Identifiability of models for clusterwise linear regression. *J. Classification* 17, 273–296.
- Hui, F.K., Warton, D.J., Foster, S.D., et al., 2015. Multi-species distribution modeling using penalized mixture of regressions. *Ann. Appl. Stat.* 9, 866–882.
- Hunter, D.R., Li, R., 2005. Variable selection using MM algorithms. *Ann. Statist.* 33.
- Ingrassia, S., Rocci, R., 2007. Constrained monotone EM algorithms for finite mixture of multivariate Gaussians. *Comput. Statist. Data Anal.* 51, 5339–5351.
- Ingrassia, S., Rocci, R., 2011. Degeneracy of the EM algorithm for the MLE of multivariate Gaussian mixtures and dynamic constraints. *Comput. Statist. Data Anal.* 55, 1714–1725.
- Izenman, A.J., 2008. *Modern Multivariate Statistical Techniques*. Springer, New York.
- Jiang, W., Tanner, M.A., 1999. On the identifiability of mixture-of-experts. *Neural Netw.* 12, 1253–1258.
- Jones, P.N., McLachlan, G.J., 1992. Fitting finite mixture models in a regression context. *Austral. J. Statist.* 34, 233–240.
- Khalili, A., 2010. New estimation and feature selection methods in mixture-of-experts models. *Canad. J. Statist.* 38, 519–539.
- Khalili, A., 2011. An overview of the new feature selection methods in finite mixture of regression models. *J. Iranian Stat. Soc.* 10, 201–235.
- Khalili, A., Chen, J., 2007. Variable selection in finite mixture of regression models. *J. Amer. Statist. Assoc.* 102, 1025–1038.
- Khalili, A., Lin, S., 2013. Regularization in finite mixture of regression models with diverging number of parameters. *Biometrics* 69, 436–446.
- Khan, A.A., Tammer, C., Zalinescu, C., 2015. *Set-valued optimization*. Springer, Heidelberg.
- Kiefer, J., 1953. Sequential minimax search for a maximum. *Proc. Amer. Math. Soc.* 4, 502–506.
- Lange, K., 2013. *Optimization*. Springer, New York.
- Lewis, M., 2004. *Moneyball: The Art of Winning an Unfair Game*. WW Norton & Company, New York.
- Major League Baseball Players Association, 2011. Agreement 2012–2016, between major league baseball and the major league baseball players association. Attachment 46, 265–276.
- Marchi, M., Albert, J., 2013. *Analyzing Baseball Data with R*. CRC Press, Boca Raton, Florida.
- McLachlan, G.J., Peel, D., 2000. *Finite Mixture Models*. Wiley, New York.
- Nelder, J.A., Mead, R., 1965. A simplex algorithm for functional minimization. *Comput. J.* 7, 308–313.
- Nguyen, H.D., McLachlan, G.J., 2015. Maximum likelihood estimation of Gaussian mixture models without matrix operations. *Adv. Data Anal. Classif.* 9, 371–394.
- Nguyen, H.D., McLachlan, G.J., 2016. Laplace mixture of linear experts. *Comput. Stat. Data Anal.* 93, 177–191.
- Quandt, R.E., 1972. A new approach to estimating switching regressions. *J. Amer. Statist. Assoc.* 67, 306–310.
- Razaviyayn, M., Hong, M., Luo, Z.-Q., 2013. A unified convergence analysis of block successive minimization methods for nonsmooth optimization. *SIAM J. Optim.* 23, 1126–1153.

- R Core Team, 2015. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria, URL <https://www.R-project.org/>.
- Schwarz, G., 1978. Estimating the dimensions of a model. *Ann. Statist.* 6, 461–464.
- Scully, G.W., 1974. Pay and performance in major league baseball. *Amer. Econ. Rev.* 64, 915–930.
- Sriperumbudur, B.K., Lanckriet, G.R., 2012. A proof of convergence of the concave-convex procedure using Zangwill's theory. *Neural Comput.* 24 (6), 1391–1407.
- Stadler, N., Bühlmann, P., van de Geer, S., 2010. l_1 -penalization for mixture regression models. *TEST* 19, 209–256.
- Tibshirani, R., 1996. Regression shrinkage and selection via the Lasso. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 58, 267–288.
- Venables, W.N., Ripley, B.D., 2002. *Modern Applied Statistics with S*, fourth ed. Springer, New York, ISBN: 0-387-95457-0, URL <http://www.stats.ox.ac.uk/pub/MASS4>.
- White, H., 1982. Maximum likelihood estimation of misspecified models. *Econometrica* 1–25.
- Wu, T.T., Lange, K., 2008. Coordinate descent algorithms for LASSO penalized regression. *Ann. Appl. Stat.* 2, 224–244.
- Yuan, M., Lin, Y., 2006. Model selection and estimation in regression with grouped variables. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 68, 49–67.
- Zhou, H., Lange, K., 2010. MM algorithms for some discrete multivariate distributions. *J. Comput. Graph. Statist.* 19, 645–665.